COVID Evidence update and opinion no 2 CT 2 5/4

Pathophysiology: current controversies

3 potential pathologies:

1. ALTERED PERFUSION RESULTING IN INAPPROPRIATE PERFUSION TO VENTILATED LUNG UNITS

Good lung compliance but poor oxygenation: CT scan showing *hyperperfusion* in gasless areas, implying impaired HPV, and therefore high shunt (Gattanoni in press).

Alveoli themselves mostly not bad: which means High Stretch maneuvers may be harmful: excess PEEP, APRV, HFOV. These could exacerbate the risk of ARDS

Previous paper showed altered lung vascular reactivity in hypertensive patients – possible link with why these patients are more likely to end up with us. ACE inhibitor use the other factor, this is being debated.

<u>Clinical implications</u>

> Following recruitment, low TV ventilation (6 mls/kg) and PEEP 5 – 8: gentle! This group may need TV of 8 mls/kg to prevent derecruitment.
>
> Consider inhaled prostacyclin to preferentially open up perfusion in the ventilated lung units

2. MICRO THROMBI IN PULMONARY VASCULATURE

Hypercoaguable state leading to V/Q mismatch from micro thrombi. Still being discussed, likely to be a factor for some patients. One paper showed no difference overall, but benefit in a subgroup of patients with high SIC score and high d-dimer. Outcome however was worse in non – high d-dimer patients.

<u>Clinical implications</u>

> Monitor d-dimer, platelet count, and coagulation. High d-Dimer - anticoagulate; low d-dimer - prophylaxis. However currently all patients are being given 0.75 mg/kg clexane unless contraindication, this remains under review. RRT potentially need full anticoagulation.

3. ARDS

> Low compliance, lung injury. Standard management still holds.

LARYNGEAL OEDEMA

Proving a problem in intubation and potentially extubation

Clinical implications

Intubation: Consider prior to induction so prepare e.g. smaller tune available

If this is contributing to respiratory distress, could result in quick resolution of respiratory difficulties once intubated

Extubation: Inspection / leak test prior to extubation
Consider dexamethasone (check infection status)
Consider therapies post extubation

FLUID RESUSCITATION

May need fluids to start with particularly if renal function shows signs of AKI. As things stabilise, can then aim for neutral / dry depending on clinical condition

Non ventilated patients

NC

Standard therapy for hypoxemic patients.

CPAP

Should be helpful. COVID-19 ARDS appears to be unusual in that compliance is good, therefore work of breathing should be less than expected. Other countries using hoods rather than masks. Unlikely to aerosolise significantly. Consider as first line if patient not coping on nasal prongs and not for intubation.

HFNC

Differing opinions. Will use up oxygen quickly, so may not be helpful if oxygen supplies limited. Concern about aerosolizing but evidence to date has not confirmed this. Less velocity than a cough, less infection risk than intubation. On balance worth considering if NC not sufficient. Guidance varies however – not recommended by ICS for example.

NIV (BIPAP)

Probably not as good an option as CPAP, HFNC or invasive ventilation, should the patient be a candidate. Given patients may have good compliance, may increase volutrauma c/f CPAP. Possible increased aerosolisation.

EARLY INTUBATION

Thought initially to be a good plan but patient then needs much higher resource. Threshold has since become higher, but **still needs to be considered earlier** than other diseases. Reasonable as long as beds allow, but then as numbers increase, inevitably patients will receive higher levels of non-invasive support prior to intubation. The argument for better control of viral shredding is not convincing once you have already have a cohort ward in place. The greatest aerosol risk is during intubation. Trial of CPAP is now taking a bigger role pre-intubation.

FLUID RESUSCITATION

May need fluids to start with particularly if renal function shows signs of AKI. As things stabilise, can then aim for neutral / dry depending on clinical condition

Drugs

Currently there is a push to only give antivirals as part of a randomised trial.

CHLOROQUINE / HYDROCHLOROQUININR

Good biological plausibility as an antiviral. Acts on ACE-2 spike binding and also increases the pH of endosomes, interfering with viral fusion and replication.. Some evidence in both animal and human data. Lacks RCT. Worth considering if it becomes available. Potentially for use in earlier stages.

VITAMIN C

The usual debate about its worth. Is used in some countries, varies according to institution. Probably not helpful, but could be considered as part of a rescue package.

REMDESIVIR

Nucleotide analogue that interferes with viral replication. Not widely available, In vitro and animal data show potentially helpful. Likely to be available through trials only.

LOPINAVIR / RITONAVIR

Used in combination usually as an HIV treatment. Chinese RCT no significant benefit v standard care but some aspects of the trial still leave the door slightly open. Watch this space.

TOCILIZUMAB

IL-6 Blocker. Has potential particularly for cytokine storm phase, but early days yet. Potentially part of a rescue package.

STEROIDS

Not for general use: no evidence of improved outcome, but results in increased viral shredding.

Could still be considered as an adjuvant in refractory septic shock

Methyl prednisone (or perhaps Dexamethasone – being discussed) if a patient suddenly deteriorates with cytokine storm / haemophagocytic syndrome is a possibility. Some evidence from China (retrospective) showed reduced mortality and quicker recovery. Notably very few on these patients were ventilated, suggesting that if patients were not for ventilation, this could be part of treatment escalation.
Potentially part of a rescue package.